of the *linear plant dynamics and quadratic performance measure* we obtain the closed-form results given in Eqs. (3.10-16) through (3.10-20a).

The reader may have noticed that the control problem of Section 3.5 is of the linear regulator type. Why then are not the optimal controls in the rightmost columns of Tables 3-2 and 3-3 linear functions of the state values? The answer is that the quantized grid of points is very coarse, causing numerical inaccuracies. When the quantization increments are made much smaller, the linear relationship between the optimal control and state values is apparent; this effect is illustrated in Problems 3-14 through 3-17 at the end of the chapter.

Another important characteristic of the linear regulator problem is that if the system (3.10-1) is completely controllable† and time-invariant, $\mathbf{H} = \mathbf{0}$, and $\mathbf{R}$ and $\mathbf{Q}$ are constant matrices, then the optimal control law is time-invariant for an infinite-stage process; that is

$$\mathbf{F}(N - K) \longrightarrow \mathbf{F} \text{ (a constant matrix)} \quad \text{as} \quad N \longrightarrow \infty.$$

From a physical point of view this means that if a process is to be controlled for a large number of stages the optimal control can be implemented by feedback of the states through a configuration of amplifier-summers as shown in Fig. 3-8(b), but with *fixed* gain factors. One way of determining the constant $\mathbf{F}$ matrix is to solve the recurrence relations for as many stages as required for $\mathbf{F}(N - K)$ to converge to a constant matrix.

Let us now conclude our consideration of the discrete linear regulator problem with the following example.

Example 3.10-1. The linear discrete system

$$\mathbf{x}(k + 1) = \begin{bmatrix} 0.9974 & 0.0539 \\ -0.1078 & 1.1591 \end{bmatrix} \mathbf{x}(k) + \begin{bmatrix} 0.0013 \\ 0.0539 \end{bmatrix} u(k) \tag{3.10-22}$$

is to be controlled to minimize the performance measure

$$J = \tfrac{1}{2} \sum_{k=0}^{N-1} [0.25x_1^2(k) + 0.05x_2^2(k) + 0.05u^2(k)]. \tag{3.10-23}$$

Determine the optimal control law.

Equations (3.10-19), (3.10-21), and (3.10-20a) are most easily solved by using a digital computer with $\mathbf{A}$ and $\mathbf{B}$ as specified in Eq. (3.10-22),

† The discrete system of Eq. (3.10-1) with $\mathbf{A}$ and $\mathbf{B}$ constant matrices is completely controllable if and only if the $n \times mn$ matrix

$$[\mathbf{B} \,\vdots\, \mathbf{AB} \,\vdots\, \dots \,\vdots\, \mathbf{A}^{n-1}\mathbf{B}]$$

is of rank n. For a proof of this theorem, see [P-2].

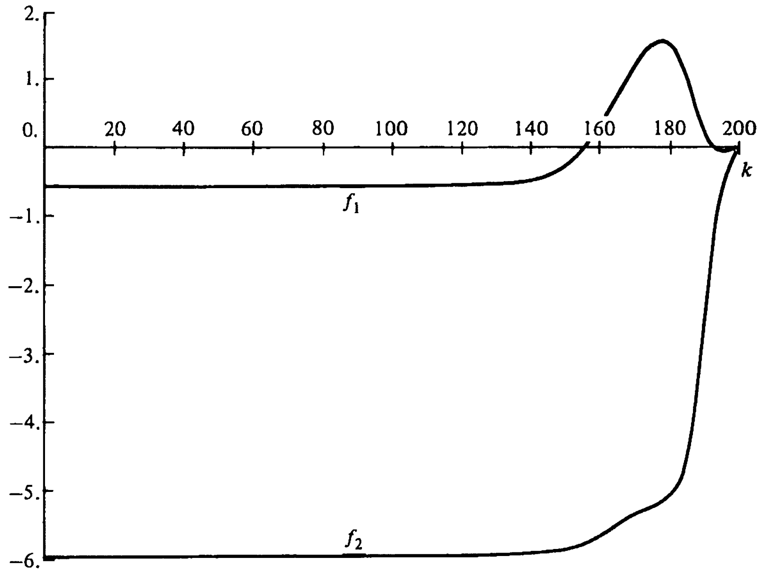

Figure 3-9(a) Feedback gain coefficients for optimal control of a second-order discrete linear regulator

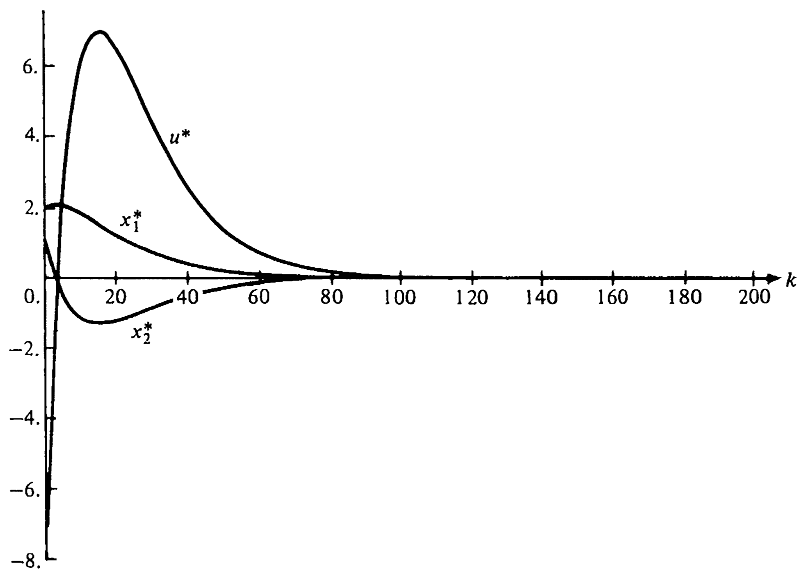

Figure 3-9(b) Optimal control and trajectory for a second-order discrete linear regulator

$$\mathbf{H} = \mathbf{0}, \qquad \mathbf{Q} = \begin{bmatrix} 0.25 & 0.00 \\ 0.00 & 0.05 \end{bmatrix}, \quad \text{and} \quad R = 0.05.$$

The optimal feedback gain matrix $\mathbf{F}(k)$ is shown in Fig. 3-9(a) for $N = 200$. Looking backward from $k = 199$, we observe that at $k \approx 130$ the $\mathbf{F}(k)$ matrix has reached the steady-state value

$$\mathbf{F}(k) = [-0.5522 \quad -5.9668], \qquad 0 \leq k \lesssim 130. \tag{3.10-24}$$

The optimal control history and the optimal trajectory for $\mathbf{x}(0) = [2 \quad 1]^T$ are shown in Fig. 3-9(b). Notice that the optimal trajectory has essentially reached $\mathbf{0}$ at $k = 100$. Thus, we would expect that insignificant performance degradation would be caused by simply using the steady-state value of $\mathbf{F}$ given in (3.10-24) rather than $\mathbf{F}(k)$ as specified in Fig. 3-9(a).

3.11 THE HAMILTON-JACOBI-BELLMAN EQUATION

In our initial exposure to dynamic programming, we approximated continuously operating systems by discrete systems. This approach leads to a recurrence relation that is ideally suited for digital computer solution. In this section we shall consider an alternative approach which leads to a nonlinear *partial* differential equation—the Hamilton-Jacobi-Bellman (H-J-B) equation. The derivation that will be given in this section parallels the development of the functional recurrence equation (3.7-18) in Section 3.7.

The process described by the state equation

$$\dot{\mathbf{x}}(t) = \mathbf{a}(\mathbf{x}(t), \mathbf{u}(t), t) \tag{3.11-1}$$

is to be controlled to minimize the performance measure

$$J = h(\mathbf{x}(t_f), t_f) + \int_{t_0}^{t_f} g(\mathbf{x}(\tau), \mathbf{u}(\tau), \tau)\, d\tau, \tag{3.11-2}$$

where h and g are specified functions, t_0 and t_f are fixed, and τ is a dummy variable of integration. Let us now use the *imbedding principle* to include this problem in a larger class of problems by considering the performance measure

$$J(\mathbf{x}(t), t, \underset{t \leq \tau \leq t_f}{\mathbf{u}(\tau)}) = h(\mathbf{x}(t_f), t_f) + \int_{t}^{t_f} g(\mathbf{x}(\tau), \mathbf{u}(\tau), \tau)\, d\tau, \tag{3.11-3}$$

where t can be any value less than or equal to t_f, and $\mathbf{x}(t)$ can be any admissible state value. Notice that the performance measure will depend on the